USGS 3DEP LiDAR Mission Parameters (2018-2020)

Governed by Lidar Base Specification v2.0 (2018) and v2.1 (September 2019)

1. Governing Specification

Lidar Base Specification (LBS) v2.0 published 2018, updated to v2.1 September 2019. LAS format version 1.4-R13 (later updated to R15). Accuracy standards per ASPRS Positional Accuracy Standards for Digital Geospatial Data (2014). Minimum quality level for all 3DEP collections: QL2.

2. Primary Instruments

2.1 Leica ALS80 (most widely used)

Laser: Nd:YAG solid-state, 1064 nm (near-infrared)
Pulse repetition rate: up to 1.0 MHz (HP/CM variants), 500 kHz (HA)
Scan rate: up to 200 Hz (HP/CM), 100 Hz (HA)
Field of view: up to 75 degrees, programmable
Max AGL: 4500 m (HP), 5000 m (UP)
Returns: unlimited discrete
Scan patterns: sine, triangle, raster (user-selectable)
GNSS: NovAtel OEM638 multi-constellation (GPS/GLONASS/Galileo/BeiDou/SBAS)
IMU: CUS-6 tactical-grade
Multiple Pulse in Air (MPiA): yes
Beam divergence: ~0.22 mrad

2.2 Other sensors used on 3DEP projects

Riegl VQ-780i: 1064 nm, polygon mirror scanner. Used in Alaska (Kuskokwim Delta).
Riegl VQ-1560i: dual-channel, up to 2.4 MHz per channel, 1064 nm.
Optech Galaxy T2000: 1064 nm, up to 2 MHz PRR. Used in Colorado (San Juan Mountains).
Leica SPL100: 532 nm single-photon lidar, wide-area mapping. Under USGS evaluation; used in South Dakota.
Optech Titan: multispectral (1550/1064/532 nm). Research and specialty applications.

3. Quality Level Specifications

3.1 Pulse spacing and density (LBS Table 1)

QL0: ANPS ≤ 0.35 m, ANPD ≥ 8.0 pls/m^2, DEM cell 0.5 m
QL1: ANPS ≤ 0.35 m, ANPD ≥ 8.0 pls/m^2, DEM cell 0.5 m
QL2 (3DEP minimum): ANPS ≤ 0.71 m, ANPD ≥ 2.0 pls/m^2, DEM cell 1.0 m
QL3: ANPS ≤ 1.41 m, ANPD ≥ 0.5 pls/m^2, DEM cell 2.0 m

Assessed against single-swath first-return data in the usable center ~95% of each swath.

3.2 Absolute vertical accuracy (LBS Table 4)

QL0: NVA $RMSE_z$ ≤ 0.050 m
QL1: NVA $RMSE_z$ ≤ 0.100 m
QL2: NVA $RMSE_z$ ≤ 0.100 m
QL3: NVA $RMSE_z$ ≤ 0.200 m

Checkpoint surveys must be 3x more accurate than expected NVA.

3.3 Relative vertical accuracy (LBS Table 2)

QL0: smooth surface $RMSD_z \leq 0.03$ m, swath overlap $RMSD_z \leq 0.04$ m
QL1/QL2: smooth surface $RMSD_z \leq 0.06$ m, swath overlap $RMSD_z \leq 0.08$ m
QL3: smooth surface $RMSD_z \leq 0.12$ m, swath overlap $RMSD_z \leq 0.16$ m

4. Collection Requirements

Season: leaf-off preferred; leaf-on requires prior USGS-NGP approval.
Returns: minimum 3 discrete returns per pulse.
Intensity: 16-bit linearly scaled per LAS 1.4 spec.
Data voids: none acceptable within a single swath. A void is any area $\geq (4 \times \text{ANPS})^2$.
Project area: Area of Interest (AOI) plus 100 m buffer = Defined Project Area (DPA).
Snow: snow-free conditions required.
Ground penetration must be adequate to produce an accurate bare-earth surface for the prescribed QL.

5. Platform and Navigation

Aircraft: fixed-wing (e.g. Cessna Caravan 208B, twin-engine turboprops). Sensor belly-mounted or pod-mounted.
GNSS: dual-frequency, multi-constellation receivers with ground base station(s).
IMU/INS: tactical-grade inertial measurement unit (e.g. CUS-6). Integrated GNSS/IMU solution provides position (lat/lon/alt) and attitude (roll/pitch/heading).

6. Typical QL2 Flight Parameters (project-dependent)

AGL: 1400-2400 m
Airspeed: 100-130 knots
FOV: 30-40 degrees half-angle
Swath width: 750-1500 m
Swath overlap: ≥ 10% (often 30-60%)
Pulse rate: 200-400 kHz
Beam divergence: 0.22-0.30 mrad
Footprint diameter: 0.25-0.50 m at typical AGL
Pulse duration: ~2.5 ns
Scan frequency: 40-70 Hz

7. Data Format and Deliverables

Point cloud format: LAS 1.4 (R13, later R15). Point Data Record Format 6, 7, 8, 9, or 10.
GPS time: Adjusted GPS Time (satellite GPS minus 1×10^9).
Waveform data: external .wdp auxiliary files.
Intensity: 16-bit normalized.
File Source ID: 0 for tiled LAS files.
Distribution format: LAZ (lossless compressed LAS).
DEM format: GeoTIFF. Null value: -999999.
QL2 DEM cell size: 1 m (or 2 US survey feet).

Hydro-flattened water surfaces and breaklines required.

8. Minimum Classification Scheme (LBS Table 5)

Class 1: Processed, but unclassified
Class 2: Bare earth (ground)
Class 7: Low noise
Class 9: Water
Class 17: Bridge deck
Class 18: High noise
Class 20: Ignored ground (near breaklines/overhangs)
Class 21: Snow (if present and identifiable)
Class 22: Temporal exclusion (nonfavored intertidal data)
No points may remain in Class 0 unless flagged as withheld. Additional classes (vegetation, buildings, wires, etc.) may be added per project.

9. Coordinate Reference System

Horizontal datum: NAD83 (various realizations). Projection: UTM or State Plane.
Vertical datum: NAVD88. Geoid model: GEOID12b (LBS v2.0), updated to GEOID18 (LBS v2.1, Sep 2019).
CRS encoding: compound WKT (COMPD_CS wrapper). All elements require EPSG AUTHORITY tags except the compound CRS itself.
Geoid name appended to VERT_CS name field. No user-defined entities (e.g. GEOID_MODEL[]) allowed.

10. Land Cover Classes for Accuracy Reporting (LBS Table 3)

NVA (non-vegetated accuracy): Class 1 - clear/open, bare earth, low grass (sand, rock, lawns, golf courses).
VVA (vegetated vertical accuracy): Class 2 - urban areas. Class 3 - tall grass/crops. Class 4 - brush/short trees. Class 5 - forested areas.
Not assessed: Class 6 - sawgrass. Class 7 - mangrove/swamps.

11. Sources

USGS Lidar Base Specification Online:
usgs.gov/ngp-standards-and-specifications/lidar-base-specification-online
USGS 3DEP Quality Levels: usgs.gov/3d-elevation-program/topographic-data-quality-levels-qls
USGS LBS Collection Requirements:
usgs.gov/ngp-standards-and-specifications/lidar-base-specification-collection-requirements
Leica ALS80 product specifications: Leica Geosystems / Hexagon
LIDAR Magazine sensor survey articles (2018-2019)